

Further investigation into the mechanisms of asphaltene deposition and permeability impairment in porous media using a modified analytical model



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HIGHLIGHTS

- New mechanism was proposed for asphaltene deposition and permeability reduction.
- Existing mechanism(s) was modified based on experimental results.
- Existing analytical model was modified to better match the experimental data.
- The modified model predicts the experimental data more accurately.

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ABSTRACT

Asphaltene deposition in porous media and subsequent in situ permeability impairment is one of the main issues during oil production which damages the reservoir rock by clogging pore throats and pore bodies as well as by changing wettability of the formation. The major mechanism(s) under which asphaltene particles deposit and impair permeability are still under investigation as an active research topic in the literature. Recently, several dynamic and static asphaltene deposition tests were conducted by Kord et al. [4] in which live oil and core samples from target reservoirs were used. According to the experimental results associated with the coreflood tests, “surface deposition” was found to be the main mechanism of permeability reduction in porous media. In addition, all the experiments confirmed that “pore throat plugging” caused linear reduction of core permeability versus time up until the “pore throat opening” mechanism comes into effect in restoring core permeability and/or flattening out the decline rate associated with permeability versus time due to flow dynamics effects. Using a novel definition of “critical velocity in pore throats”, this new mechanism was mathematically modeled. In order for precise demonstration and quantification of the “surface deposition” mechanism, a novel experimental apparatus was designed and constructed by Soulgani et al. [26] to measure mass of deposited asphaltene versus time. The results were used to present a new term describing the “surface deposition” mechanism. A three-phase, four-component black-oil simulator was then developed and coupled with the deposition model including two mechanisms of “surface deposition” and “pore throat opening”. Subsequently, the simulator was verified using two sets of experimental results. A very good match between the values predicted by the model and those originated from the experiments were observed.

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1. Introduction

Asphaltene deposition causes challenging operational/reservoir management difficulties including permeability reduction, changing the rock wettability and increasing the fluid viscosity. Developing a predictive tool could play a vital role in estimating

the amount and location of asphaltene damage and subsequent treatment operations. However, developing physical and mathematical models to precisely describe deposition mechanisms and capture the true physics of deposition and impairment is still at initial stages of development.

Changes in pressure, temperature or composition would cause stabilized asphaltenes to precipitate from the oil solution [1]. Various models with different focal points have been developed using reversible and irreversible thermodynamics to model asphaltene

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Nomenclature

E_A	volume fraction of deposited asphaltene, dimensionless	ρ_o	density of oil phase, g/cm ³
E_a	attachment activation energy, J/mol	ρ_A	density of asphaltene, g/cm ³
R_{Aso}	volume ratio of soluble asphaltene in the oil phase, dimensionless	α	surface deposition rate coefficient, 1/s
C_A	suspended asphaltene saturation, dimensionless	β	entrainment rate coefficient, 1/cm
W_{Sao}	suspended asphaltene, dimensionless	γ	instantaneous plugging deposition rate coefficient, 1/s
q_{osc}	flow rate, cm ³ /s	σ	snow-ball effect deposition constant, dimensionless
S_g	saturation of gas phase, dimensionless	ϕ	porosity, fraction
S_o	saturation of oil phase, dimensionless	ϕ_0	initial porosity, fraction
$\dot{m}_d$	rate of asphaltene deposition, kg/m ² s	k_0	initial permeability, md
u_o	velocity of oil phase, cm/s	k_d	correction factor, (m/s) ²
v	interstitial oil velocity, cm/s	f_p	permeability correction constant, dimensionless
V_C	critical interstitial velocity for entrainment, cm/s	t	time, s
V_{CP}	critical interstitial velocity for entrainment in modified model, cm/s	Subscripts	
V_{CT}	critical interstitial velocity in pore throats, cm/s	A	asphaltene
V_m	molar volumes of oil, m ³ /kmol	cr	critical
B_o	oil formation volume factor, m ³ /S m ³	o	oil
B_A	asphaltene formation volume factor, m ³ /S m ³	g	gas
R	Universal gas constant, cal/mol K	m	molar
T	temperature, K	d	deposit
ρ_g	density of gas phase, g/cm ³	sc	standard condition

precipitation. In addition, scaling equations are recently proposed in the form of simple models to predict the asphaltene precipitation [2,3]. A new class of scaling equations was developed recently by Kord and Ayatollahi [4] which is capable of predicting asphaltene precipitation from live oil at reservoir conditions.

Asphaltene precipitation due to any thermodynamic instability cannot be considered as a unique criterion for asphaltene deposition. It is the rock morphology and flow hydrodynamics which govern the extent of permeability damage [5,6]. The model proposed by Gruesbeck and Collins [7] for migration of fine particles in porous media is the base for most of the asphaltene deposition models. Ali and Islam [8] used parallel pathways idea considering surface excess theory of Sircar et al. [9] and developed a mathematical model which was verified against experimental data. Minsieux et al. [10] conducted series of coreflood experiments with different core and fluid samples at reservoir temperature in order to investigate the effect of asphaltene deposition on permeability of core samples. Chauveteau and Nabzar [11] analyzed different mechanisms leading to permeability damage due to particle deposition based on different forces acting in various situations. They characterized deposition regimes by a few non-dimensional numbers. Considering parameters such as particle and pore size, shape and roughness, flow rate, concentration, viscosity, particle–particle and particle–pore wall interaction energies and pore surface physicochemical heterogeneity, they concluded that the deposition regime determines the deposition kinetics. Using Carman–Kozeny equation, Nghiem et al. [12] proposed a compositional simulator, which was coupled with an empirical linear correlation for the induced-damage factor.

Based on the Ali and Islam deposition model [8], Kocabas et al. [13] developed a near wellbore model which considers both mechanical trapping and adsorption which was analytically solved in the case of no adsorption and was then compared against some sets of experimental data. Almeida et al. [14] developed a single well model to simulate asphaltene precipitation, deposition and well plugging during natural depletion. Based on this idea, Solaimany-Nazar and Zonnouri [15] used the same approach to examine the effect of production rate and initial permeability on asphaltene deposition behavior around the wellbore.

Zekri and Shedid [16] investigated the effect of fracture characteristics on its permeability reduction by asphaltene deposition. A single fracture with different properties (i.e. porosity, permeability and fracture angle), but with constant aperture was used to investigate the effect of different parameters on permeability damage. Moreover, the effect of asphaltene precipitation on the end point relative permeability values at S_{wi} was also studied in some experimental work [17,18].

Danesh et al. [19] presented three mechanisms contributing to entrapment of suspended asphaltene particles through porous media: surface deposition, entrainment, and pore-throat plugging. Wang and Civan [20,21] suggested a three-parameter equation for deposition rate of asphaltenes as a function of the mechanisms introduced by Danesh et al. [19], along with Gruesbeck and Collins [7] model. They also proposed a mass balance equation to calculate the amount of asphaltene ready for deposition. The whole sets of equations were incorporated into a three-dimensional, three-phase black oil simulator and were solved by finite-difference method. Surface deposition, entrainment, and pore throat plugging rate coefficients were estimated by regression analysis of the experimental data provided by Minsieux et al. [10]. Papadimitriou et al. [22] also performed some experiments to investigate the asphaltene deposition process and its subsequent permeability impairment in core samples under stimulant and realistic flow conditions.

Mendoza et al. [23] conducted coreflood experiments to investigate the processes of asphaltene precipitation and deposition during natural depletion and their effects on core permeability. Moreover, they developed a mathematical model based on the transport of stable particulate suspensions in porous media and validated it with experimental data. In their proposed model, asphaltene adsorption and trapping were considered as main causes of deposition. Lawal and Vesovic [24] implemented a simple one-parameter semi-analytic model for asphaltene-induced reduction of porosity, pore-radius and permeability in closed systems. They considered the sedimentation as the dominant deposition mechanism to demonstrate dynamics of asphaltene deposition under stagnant conditions. Results of this study showed that the damage rate depends on the size of asphaltene aggregates as well as on the rock and fluid properties.

Clear understanding of different aspects of asphaltene deposition can help the petroleum industry to design proper treatment operations [25]. Physical explanation of the deposition process is still at initial stages of its development. To further investigate the mechanisms involved in the impairment process, Soulgani et al. [26] modified Wang and Civan [20,21] model by presenting a new expression for the surface deposition term and implementing this expression in the asphaltene deposition modeling. The results of their experiments showed that increasing temperature as well as increased concentration of precipitated asphaltenes caused the surface deposition rate to increase. Moreover, the results of the experiments showed that the rate of surface deposition is inversely proportional to oil velocity.

To physically recognize and quantify different aspects of asphaltene deposition, Kord et al. [27] designed and conducted several static and dynamic asphaltene deposition tests using both stock tank oil and live reservoir fluids and core samples from target reservoir. From these experiments, it was found that surface deposition of asphaltene particles is the main mechanism of formation damage in porous media and that the resulting permeability impairment obeys an exponential behavior. All the experiments confirmed that permeability of all the tested cores reduced linearly versus pore volume injected (i.e. process time) under the pore throat plugging mechanism of permeability impairment until a counterbalancing mechanism of pore throat opening comes into effect.

In this paper, the impairment model proposed by Wang and Civan [20,21] is modified based on these two latter studies discussed above; surface deposition and the counterbalancing mechanism of pore throat opening is discussed and further incorporated into the Wang and Civan [20,21] impairment model. The proposed modified model is then used to predict permeability impairment during coreflood experiments. The model predictions are in excellent agreement with the experimental data obtained by Kord et al. [27].

2. Modeling of asphaltene deposition and subsequent permeability impairment

In order to modify reservoir performance based on the effect of asphaltene deposition and subsequent permeability reduction, it is essential to include different deposition mechanisms into reservoir simulation. In this study, a modified three-phase four-component black oil simulator is being introduced based on the model originally proposed by Wang and Civan [20,21] and the experimental and modeling work presented earlier by the authors in Ref. [27] of this paper.

It is of great importance to differentiate between the static and dynamic concentration of asphaltenes ready for precipitation and deposition. Under different thermodynamic and hydrodynamic conditions, asphaltenes may precipitate out of solution but only a portion of the precipitated asphaltenes may deposit on rock surfaces. In order to take into account the dynamic asphaltene concentration (i.e. concentration of asphaltene capable of deposition on rock surfaces), a material balance equation for asphaltene component was proposed by Wang and Civan [20,21]:

$$-\nabla \cdot \left(\frac{R_{Ao} u_o}{B_o} + \frac{C_A u_o}{B_A} \right) - \frac{C_A q_{osc}}{B_A} + \frac{R_A q_{osc}}{B_o} = \frac{\partial}{\partial t} \left(\frac{C_A \phi S_o}{B_o} + \frac{R_A \phi S_o}{B_o} + \frac{E_A}{B_A} \right) \quad (1)$$

in which C_A is the suspended asphaltene saturation which is defined as:

$$C_A = \frac{\rho_o}{\rho_A} w_{SAo} \quad (2)$$

Eq. (1) is one of the main equations that form the deposition model, but estimating the volume fraction of deposited asphaltenes is another important factor to be considered. Wang and Civan [20] presented the following equation to calculate the rate by which the volume fraction of deposited asphaltenes changes based on different mechanisms involved in the deposition process:

$$\frac{dE_A}{dt} = \alpha C_A \phi - \beta E_A \left(\frac{u_o}{\phi} - V_c \right) + \gamma (1 + \sigma E_A) u_o C_A \quad (3)$$

In this equation, the first, second and third terms represent surface deposition of asphaltene particles, entrainment of deposited particles, and the pore throat plugging rate, respectively. The core part of the model is five tuning parameters: α is the surface deposition rate constant, β is the entrainment of deposits constant, V_c is the critical interstitial velocity for entrainment, γ is the pore throat plugging rate constant, and σ is the snowball rate constant.

Porosity is reduced by the amount of fractional pore volume occupied by deposited asphaltene particles. Hence, instantaneous porosity and local permeability impairment could be given by:

$$\phi = \phi_0 - E_A \quad (4)$$

$$k = f_p k_0 \left(\frac{\phi}{\phi_0} \right)^3 \quad (5)$$

in which ϕ_0 is the initial porosity; k_0 is the initial permeability and f_p is the permeability correction constant.

3. Modification of Wang and Civan deposition model [20,21] focusing on novel definitions for “surface deposition” and “Pore Throat Plugging Rate” mechanisms

Through extensive experimental and modeling studies presented in reference 27 of this paper, it was found that there is discrepancy between the experimental data and predictions originated from Wang and Civan [20,21] model especially at greater values of injection pore volumes. An example of such a discrepancy is presented in Fig. 1 in which a relatively considerable error in predictions of Wang and Civan deposition model [20,21] versus the experimental data can be observed. For each particular impairment test, the general rock properties as well as the operating conditions are provided in Table 1. In addition, some data about the reservoir fluid properties at in-situ conditions are provided in Table 2. The data associated with SARA analysis of different oils as well as their compositional characteristics are provided in Tables 3 and 4.

As it is demonstrated in Fig. 1, the trapping mechanisms incorporated into the Wang and Civan deposition model [20,21] cannot fully capture the entire plugging regimes identified in the conducted experiments. The experimental results presented in Ref. [27] of this paper could elucidate some less understood features of asphaltene deposition and permeability impairment; hence formed the basis for developing the modeling framework presented in this paper. According to the experimental data presented elsewhere [27], four different flow regions can be recognized during the continuous flow of asphaltenic oil through porous media which results in asphaltene deposition and subsequent permeability reduction as a result of pressure drop across the cores. The permeability ratio versus process time (i.e. pore volume injected) graph can be discretized into these four consecutive flow regions as presented in Fig. 2:

Region 1: In this region, which corresponds to the initial stages of the oil flood through the core samples, both mechanisms of “surface deposition” and “entrainment”, associated with asphaltene deposition and permeability impairment, act simultaneously; however, the dominant permeability impairment mechanism is “surface deposition”. This represents itself in the form of exponen-

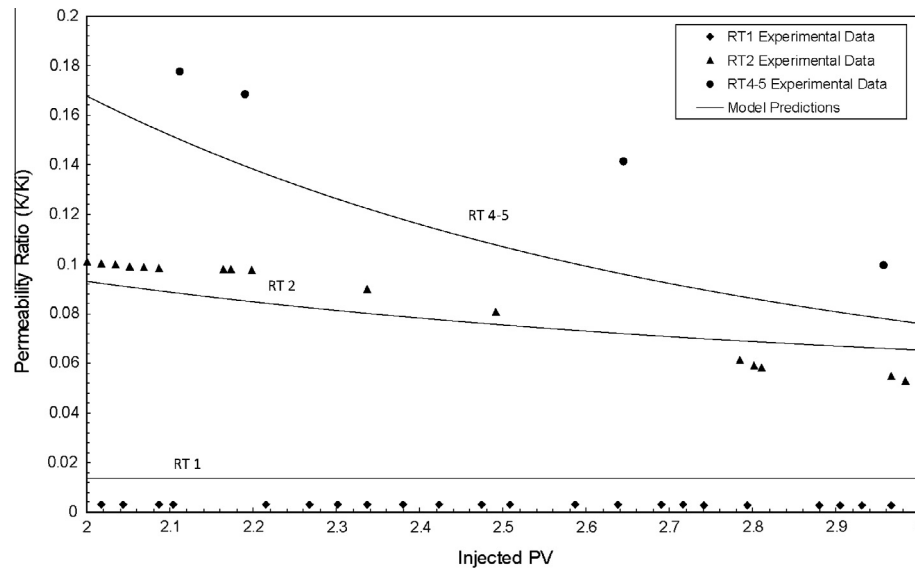


Fig. 1. Permeability impairment experimental data along with predicted values obtained from Wang and Civan plugging model [20,21] during extended time (i.e. 2–3 PV) of flow of asphaltenic oils through core samples RT1, RT2 and RT4-5 (extracted from Ref. [27]).

Table 1
General specifications of permeability impairment tests [27].

Impairment test information	RT1	RT2	RT3-1	RT4-5
Lithology	Carbonate	Carbonate	Carbonate	Carbonate
Oil type used	Oil-A	Oil-A	Oil-B	Oil-B
API gravity	29	29	21	21
Asphaltene content (wt.%)	13.5	13.5	3.75	3.75
Viscosity (cp)	5	5	2	2
Porosity (fraction)	0.046	0.07	0.254	0.1404
Permeability (md)	2.35	1.23	14.19	18
Length (cm)	8.83	8.87	8.57	8.93
Diameter (cm)	3.69	3.705	3.72	3.81
Injection flow rate (cc/min)	0.2	0.2	0.2	0.2
Temperature (°C)	54.44	54.44	87.78	87.78
Back pressure (MPa)	Variable	0.007	0.06	3.45

Table 2
Fluid properties at reservoir conditions [27].

SPECIFICATION	Unit	Oil-A	Oil-B
Reference depth	m (Sub-sea)	3750	3350
Reservoir pressure	MPa	49	41
Reservoir temperature	°C	124	106
Saturation pressure	MPa	11.873	11.087
Solution GOR	S m ³ /S m ³	80.1	67.8
Oil formation volume factor @ reservoir pressure	m ³ /S m ³	1.230	1.283
Density of oil @ reservoir pressure	g/cc	0.821	0.799
Gravity of residual oil	°API	29.17	20.75
Specific gravity of residual oil	15.6/15.6 °C	0.932	0.929
Viscosity of oil @ saturation pressure	cp	1.592	2.030

Table 3
SARA analysis [27].

Specification	Unit	Oil-A	Oil-B
Saturated	wt.%	30.1	43.67
Aromatic	wt.%	42.1	52.09
Resin	wt.%	13.36	0.49
Asphaltene	wt.%	13.5	3.75

tial decline in permeability ratio as time passes by. While the life-span of this particular region is very short, it imposes the maximum observed permeability damage to the porous media to the extent of about 50–70% reduction in initial permeability.

Region 2: In this region, the surface deposition mechanism will be more severe if the contribution of entrainment mechanism is not considered. The counter effect of these two mechanisms results in a steady-state condition in the form of almost constant permeability ratio presented in Fig. 2 by white-colored circles after about 0.4 injected PV. In this region, the rate of deposition of asphaltene particles is almost equal to the rate of detachment or entrainment which results in almost no change in permeability ratio with respect to the process time.

Region 3: After about 0.5–1 PV injection, it is observed that there is a linear declining trend in permeability ratio versus the pore volume injected (Fig. 2). This behavior is attributed to the pore throat plugging mechanism. Approximately 15–20% of permeability

Table 4

Composition of crude oil samples [27].

Component	Crude Oil-A	Crude Oil-B
H ₂ S	2.70	0.16
N ₂	0.21	0.45
CO ₂	5.14	1.55
C ₁	22.00	22.90
C ₂	7.10	7.64
C ₃	5.34	5.35
iC ₄	0.99	1.16
nC ₄	2.78	2.74
iC ₅	1.12	0.93
nC ₅	1.41	1.12
C ₆	5.55	4.27
C ₇	3.84	4.29
C ₈	4.02	3.60
C ₉	2.85	3.31
C ₁₀	2.83	2.95
C ₁₁	2.58	2.72
C ₁₂₊	29.54	34.86
Total	100	100

reduction is experienced in this region where main contributing drive is the pore throat plugging mechanism.

Region 4: After continuous injection of about 1 PV of the asphaltenic crude oil through the core sample, an exponential behavior ascribed to a so-called “pore throat opening” mechanism, i.e. a deposition retarding drive, is recognized in addition to the active deposition mechanisms. During the extended injection durations, this counter-deposition mechanism provides a retarding effect on asphaltene deposition and subsequent permeability impairment to the extent that the permeability ratio was just decreased by about 2–10% during the extended period of 1–3.5 pore volume injection.

The above mentioned regions associated with permeability impairment plot along with the dominant active mechanisms in each particular region can be observed in Refs. [10,20,21].

Investigation of the experimental works published in the open literature shows the importance of surface deposition mechanism. Results of the coreflood tests presented in Ref. [27] showed that surface deposition of asphaltene particles is the main cause for permeability reduction in the form of an exponential decrease in the permeability ratio versus the PV of oil injected. In order to further

investigate this mechanism, an experimental apparatus was constructed to obtain mass of asphaltene deposited as a function of time as was described in details in Ref. [26]. The experimental apparatus used in this work is a pipe-shaped reactor. In other words, since it is not a porous medium, all surfaces are available for deposition. As a result, porosity could not be incorporated as a parameter in the proposed equation. The mass of deposited asphaltene can be obtained using Eq. (6) in which several parameters such as temperature, velocity and precipitated asphaltene concentration (C_{AS}) are contributing in determination of the mass flux of deposited asphaltene through the surface deposition mechanism [26]:

$$\dot{m}_d = \frac{K_d}{v} e^{-\frac{E_a}{RT}} C_{AS} \quad (6)$$

It is worthwhile to note that the effect of velocity should be typically included in the mathematical term describing the entrainment mechanism; however, because the authors' intention was to amalgamate all these parameters in the form of a single equation, such an inclusion was taken into account. The main finding extracted from these experiments and the mathematical modeling work is that the term $e^{-\frac{E_a}{RT}}$ is now included in the surface deposition mechanism. Therefore, the surface deposition mechanism in porous media can be represented mathematically in the general form of $\alpha e^{-\frac{E_a}{RT}} C_A \phi$.

As was described based on the behavior of permeability reduction versus the number of injected pore volumes of the asphaltenic crude oil through the porous media, an additional mechanism was hypothesized namely as the “pore throat opening” mechanism which counterbalances the “pore throat plugging” mechanism. This newly hypothesized mechanism is responsible for the very mild reduction in permeability ratio during the extended injection durations. All these four mechanisms are schematically presented in Fig. 3. By clogging up the pore throats through the “pore throat plugging” mechanism, the pore velocity of fluid at the pore body region and also adjacent to the pore constrictions will be increased due to reduced cross sectional area available to flow as long as the head pressure at the inlet of the core package remains constant. This localized increase in the pore velocity along with the pore-scale mixing associated with the flow of particulate-containing oil through the pore structure results in entrainment of some of

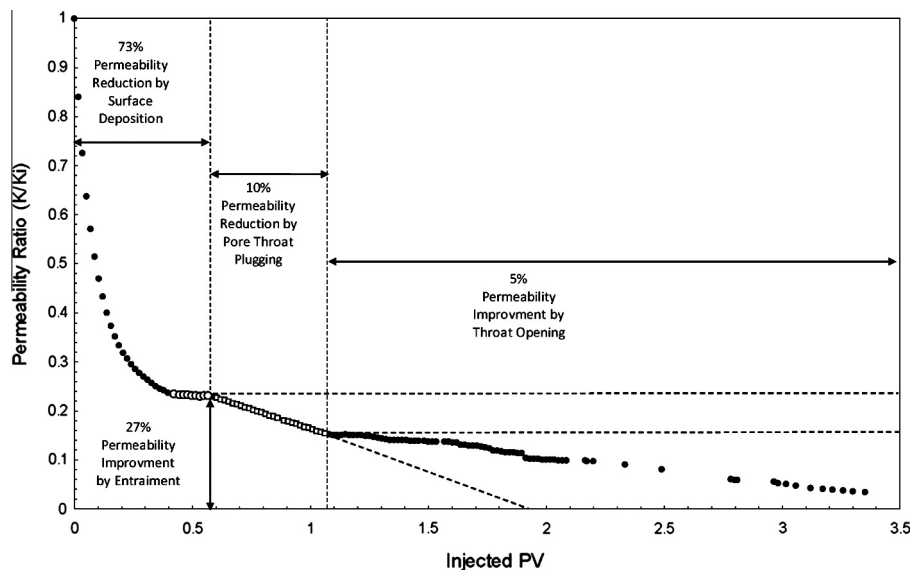


Fig. 2. Permeability impairment accompanied by approximate active regions for different deposition mechanisms during constant flow rate injection of asphaltenic crude oil through core sample RT2 (extracted from Ref. [27]).

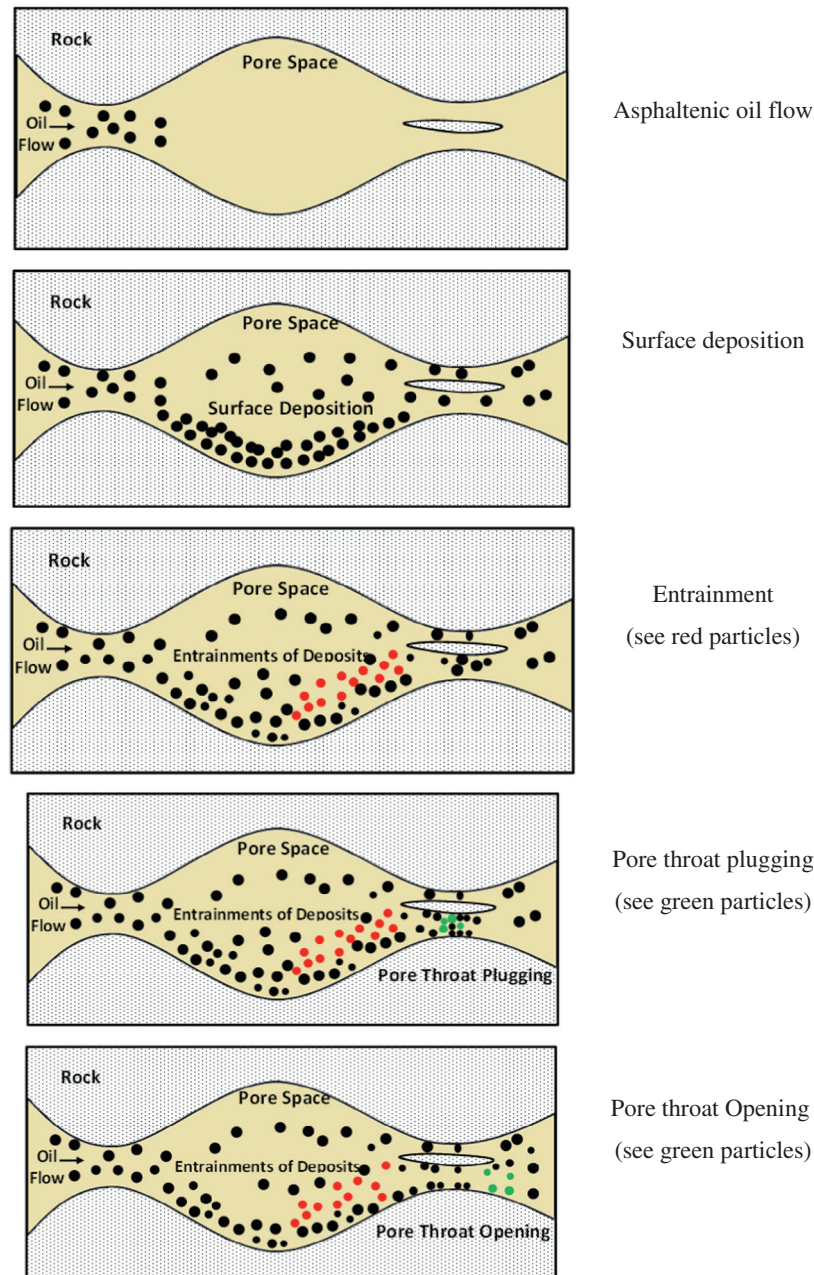


Fig. 3. Schematic of different asphaltene deposition and permeability impairment mechanisms in porous media originated from continuous flow of asphaltenic crude oil.

the already-deposited asphaltene particles on the inner active surfaces of the pore. In addition, this increase in pore velocity expels out a portion of the asphaltene particles clogging up the pore throats, hence provides less resistant flow pathways for the particulate-containing liquid to pass through the partially re-opened pore constrictions. Due to this partial re-opening of the plugged pore throats through the “pore throat opening” mechanism, one may explain why a particular porous medium never becomes totally plugged during the natural depletion process.

In order to modify the Wang and Civan model [20,21] based on this finding at extended lifespan of the natural depletion, the “pore throat opening” mechanism can be combined with the “pore throat plugging” mechanism to obtain a net value parameter controlling the rate of clogging of the pore throats during the entire life of the natural depletion process. The deposited asphaltene in a particular pore throat can be removed when the flow velocity at that particular location (i.e. pore throat) reaches the critical pore velocity

for entrainment. By introducing the critical interstitial velocity in the pore throat region into the picture and subtracting its effect from the expression representing the pore throat plugging rate, one could develop a new mathematical expression describing the net value of the “pore throat plugging” mechanism. This

Table 5

Tuning parameters for Wang and Civan deposition model [20,21] optimized to match the experimental data [27].

Matching Parameter	RT2	RT3-1	RT4-5
V_{cr} (cm/s)	0.000194806	0.009431	0.000158724
α (1/s)	0.095	0.055	0.0035
β (1/cm)	0.37	0.0	0.05
γ (1/cm)	6	0.05	0.25
σ (constant)	0	35	45
f_p (constant)	1	1	1

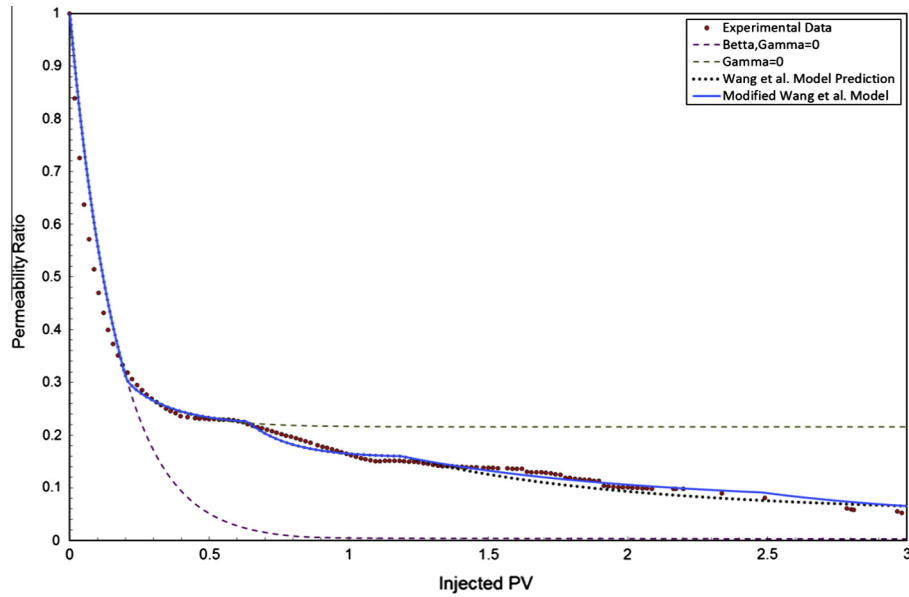


Fig. 4. Permeability impairment versus injected pore volumes of oil – model predictions versus experimental data for RT2 test.

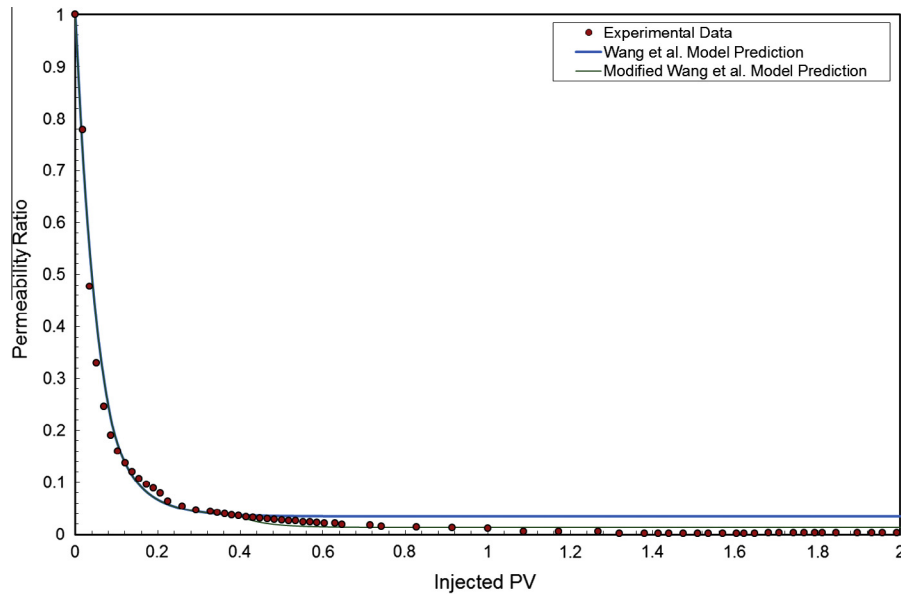


Fig. 5. Permeability impairment versus injected pore volumes of oil – model predictions versus experimental data for RT3-1 test.

mechanism takes in effect more significantly when approaching towards the late lifespan of the permeability impairment through natural depletion recovery process. The modified mathematical expression for the pore throat plugging mechanism (i.e. the net value) can be represented as:

$$\text{Pore throat plugging} = \gamma(1 + \sigma E_A)(u_o - V_{CT})C_A \quad (7)$$

Considering the new definition of surface deposition mechanism as well as the above modification regarding the pore throat opening drive, one can modify the Wang and Civan model [20,21] as follows:

$$\frac{dE_A}{dt} = \alpha e^{\frac{-E_A}{RT}} C_A \varphi - \beta E_A \left(\frac{u_o}{\varphi} - V_{CP} \right) + \gamma(1 + \sigma E_A)(u_o - V_{CT})C_A \quad (8)$$

In this equation, the first term is the new mathematical representation of the surface deposition mechanism, the second term is the entrainment of deposits which is similar to the originally

proposed term in the Wang and Civan model [20,21], and the third term shows the modified pore throat plugging rate. This modified model has six matching parameter: α is the surface deposition rate constant, β is the entrainment of deposits constant, γ is the pore throat plugging rate constant, σ is the snowball rate constant, V_{CP} is the critical interstitial velocity for entrainment of pores and V_{CT} is the critical interstitial velocity for entrainment of throats.

4. Model predictions

In order to testify whether the modified model can predict the permeability impairment more effective than the original Wang and Civan model [20,21], three of the dynamic asphaltene deposition tests under the natural depletion process were selected from the series of permeability impairment tests conducted by Kord et al. [27]. Previously, it was attempted to match the experimental

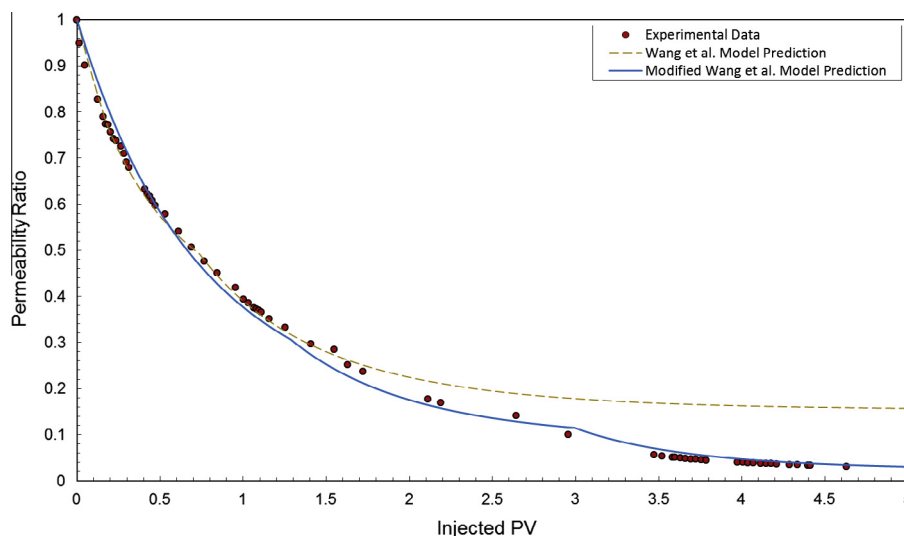


Fig. 6. Permeability impairment versus injected pore volumes of oil – model predictions versus experimental data for RT4-5 test.

Table 6

Tuning parameters for the modified Wang and Civan model [20,21] optimized to match the experimental data [27].

Matching Parameter	RT2	RT3-1	RT4-5
V_{CP} (cm/s)	0.000194806	0.009431	0.000158724
α (1/s)	0.0089	0.0061	0.00098
β (1/cm)	0.37	0.0	0.05
γ (1/cm)	6	0.5	0.25
σ	0	35	45
f_p	1	1	1
V_{CT}	0.0000175	0.000651	0.0000114

data with Wang and Civan model [20,21] by optimizing the tuning parameters [27]. The tuned parameters are shown in Table 5. Even with the best possible optimized tuning values, Wang and Civan deposition model [20,21] could not satisfactorily match the experimental results as observed in Figs. 4–6 especially during the extended process times in which the “pore throat opening” mechanism comes into effect.

In order to evaluate the modified Wang and Civan deposition model [20,21], Eq. (8) was used in order to take into account the newly developed expression for surface adsorption mechanism as well as to account for re-opening of the clogged pore throats through the so-called “pore throat opening” mechanism. The tuning parameters associated with Eq. (8) were optimized in order to provide an acceptable match to the experimental data [27] (Table 6). The predicted permeability impairment values versus the pore volume injected for these three tests using the modified deposition model are shown in Figs. 4–6. As observed in these figures, there is a very good improvement in matching the experimental data using the modified Wang and Civan [20,21] model especially during the extended injection period. Beside the improvement in predictions, this modified model can explain the pore throat plugging mechanism more realistically compared to the original Wang and Civan deposition model [20,21].

In order to verify the validity and accuracy of our modified model using a second set of permeability impairment data, the experimental results reported by Minssieux et al. [10] and modeled by Wang and Civan [20,21] were selected. It is worthwhile to mention that the impairment regions described earlier can also be noted in this set of experimental data. However due to some differences in operating conditions and fluid and rock properties, there are not as many data points in each particular region associated

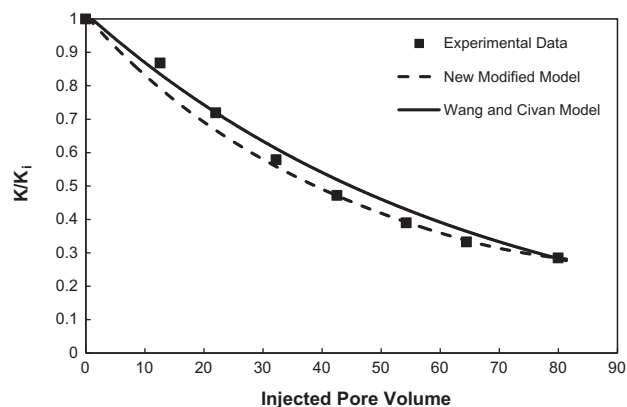


Fig. 7. Permeability ratio versus pore volume injected for GF3 sample [10] along with Wang and Civan deposition model [20,21] as well as new modified model.

with Minssieux et al. [10] impairment data as one can find in our sets of impairment data. In particular, the “pore throat opening” mechanism cannot be distinguished by processing this set of impairment data. Several reasons could be postulated such as much greater porosity and permeability values associated with their employed porous media as well as less asphaltene content

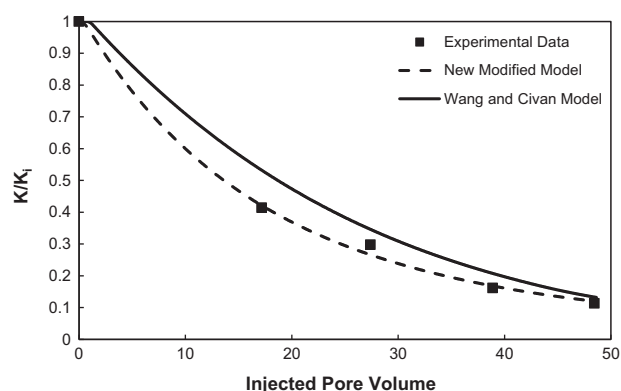


Fig. 8. Permeability ratio versus pore volume injected for GV10 sample [10] along with Wang and Civan deposition model [20,21] as well as new modified model.

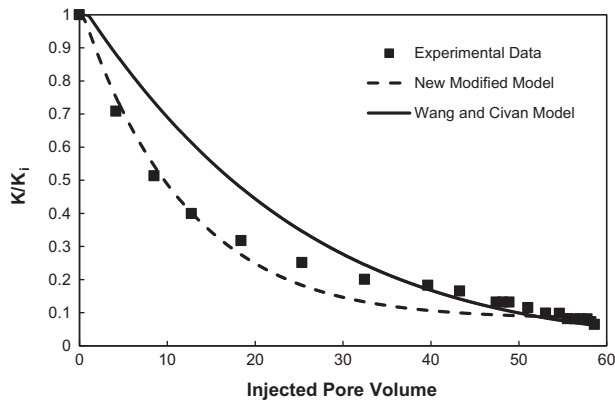


Fig. 9. Permeability ratio versus pore volume injected for GV5 sample [10] along with Wang and Civan deposition model [20,21] as well as new modified model.

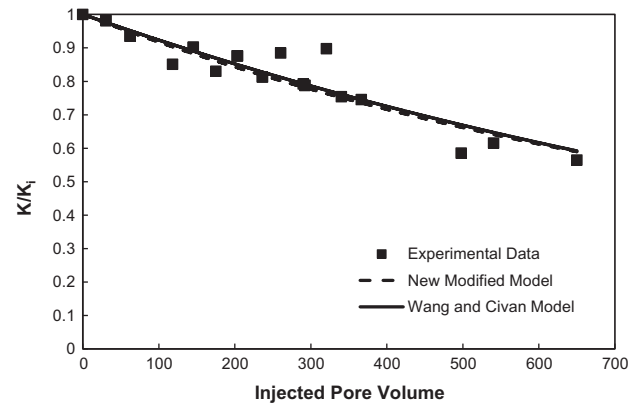


Fig. 12. Permeability ratio versus pore volume injected for HMD sample [10] along with Wang and Civan deposition model [20,21] as well as new modified model.

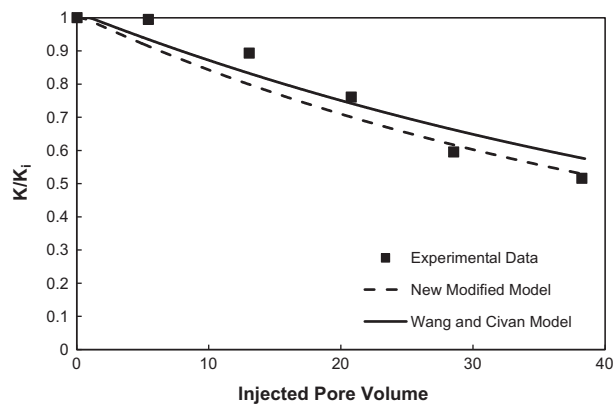


Fig. 10. Permeability ratio versus pore volume injected for GP9 sample [10] along with Wang and Civan deposition model [20,21] as well as new modified model.

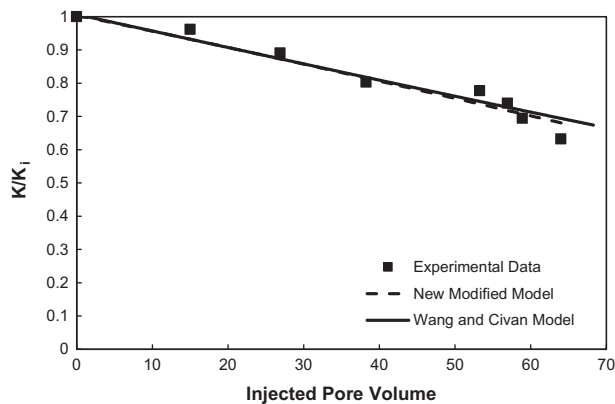


Fig. 11. Permeability ratio versus pore volume injected for GF1 sample [10] along with Wang and Civan deposition model [20,21] as well as new modified model.

of the fluid used in their study. In Figs. 7–12, these experimental data are shown along with the Wang and Civan deposition model [20,21] as well as our modified predictive model.

Considering the experimental data and predicted permeability impairment trends presented above, it is clear that our modified deposition and impairment model can provide a more realistic match for the experimental data because of the enhancement made in defining the surface adsorption mechanism as well as

incorporating the newly defined “pore throat opening” mechanism which is being active during the extended injection periods.

5. Conclusions

According to the literature review on asphaltene deposition in porous media and subsequent permeability damage modeling, it was found that Wang and Civan deposition model [20,21] is not capable of predicting consistent match values compared to the experimental results especially during the extended injection periods associated with natural depletion process of asphaltene deposition in porous media. During the late lifespan stages of oil injection, the dominant contributing mechanisms for asphaltene deposition are different than the ones presented in the literature, so are their mathematical expressions in terms of rock and fluid properties. These discrepancies could be observed more clearly in rocks with lower porosity and permeability values. Based on the results of this research, a new mechanism of “pore throat opening” is proposed. Besides, a new model proposed by Soulgani et al. [26] for surface deposition mechanism was also modified and used to better describe the asphaltene deposition mechanisms. These modified deposition models were coupled with fluid flow equations to simulate permeability reduction in coreflood tests. The results are very promising with an excellent match between the predicted values using the modified deposition model and the experimental data. The calculated damaged permeabilities are in a very good agreement with the experimental data.

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